

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 9 - Unit Test 3

Unit Test 3 - Answer Key

Class: Class 9

Assessment Type: Unit Test 3

Total Marks: 10

Duration: 30 minutes

Topics Covered: SQL

ICT Teaching Resources Portal

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Section I: MCQ (5 x 1 = 5 marks)

- Answer:** b) ALTER TABLE ... ADD
 - Explanation:** ALTER TABLE table_name ADD column_name datatype is the correct SQL command to add a new column to an existing table.
- Answer:** a) Top 3 students by marks
 - Explanation:** ORDER BY marks DESC sorts marks from highest to lowest. LIMIT 3 returns only the first 3 rows.
- Answer:** c) ORDER()
 - Explanation:** ORDER() is not an SQL aggregate function. SUM(), COUNT(), and AVG() are aggregate functions.
- Answer:** c) Missing/unknown value
 - Explanation:** IS NULL checks for NULL values, which represent missing or unknown data in SQL.
- Answer:** b) GROUP BY
 - Explanation:** GROUP BY groups rows that share common values, typically used with aggregate functions like COUNT(), SUM(), AVG().

Section II: Short Answer (5 x 1 = 5 marks)

- Write SQL queries for Products(pid, name, price, category, stock).**
 - Answer:**

```
-- Products with stock less than 10
SELECT * FROM Products WHERE stock < 10;

-- Count products in each category
SELECT category, COUNT(*) AS product_count FROM Products GROUP BY category;

-- Average price per category
SELECT category, ROUND(AVG(price), 2) AS avg_price FROM Products GROUP BY category;
```
- Differentiate between INNER JOIN and LEFT JOIN.**
 - Answer:** INNER JOIN returns only rows that have matching values in both tables. LEFT JOIN returns all rows from the left table and matching rows from the right table (NULL for non-matching right rows).

```
-- INNER JOIN: Only students with results
SELECT s.name, r.marks FROM Students s INNER JOIN Results r ON s.roll = r.roll;

-- LEFT JOIN: All students, with results if available
SELECT s.name, r.marks FROM Students s LEFT JOIN Results r ON s.roll = r.roll;
```
- Write a query to display the third highest salary.**

- **Answer:**

```
SELECT DISTINCT salary
FROM Employees
ORDER BY salary DESC
LIMIT 1 OFFSET 2;
```

- Alternative using subquery:

```
SELECT MAX(salary) FROM Employees
WHERE salary < (SELECT MAX(salary) FROM Employees
               WHERE salary < (SELECT MAX(salary) FROM Employees));
```

4. What is a Primary Key?

- **Answer:** A Primary Key is a column (or set of columns) that uniquely identifies each row in a table. It cannot contain NULL values and must be unique. A table can have only one PRIMARY KEY, but it can consist of multiple columns (composite primary key).

5. Write SQL to display duplicate records from a table.

- **Answer:**

```
-- Create a table with duplicates
CREATE TABLE Employees_dup (id INT, name VARCHAR(50), dept VARCHAR(30));
INSERT INTO Employees_dup VALUES (1, 'Amit', 'IT'), (2, 'Priya', 'HR'), (1, 'Amit', 'IT');
```

```
-- Find duplicate records
SELECT id, name, dept, COUNT(*) as count
FROM Employees_dup
GROUP BY id, name, dept
HAVING COUNT(*) > 1;
```